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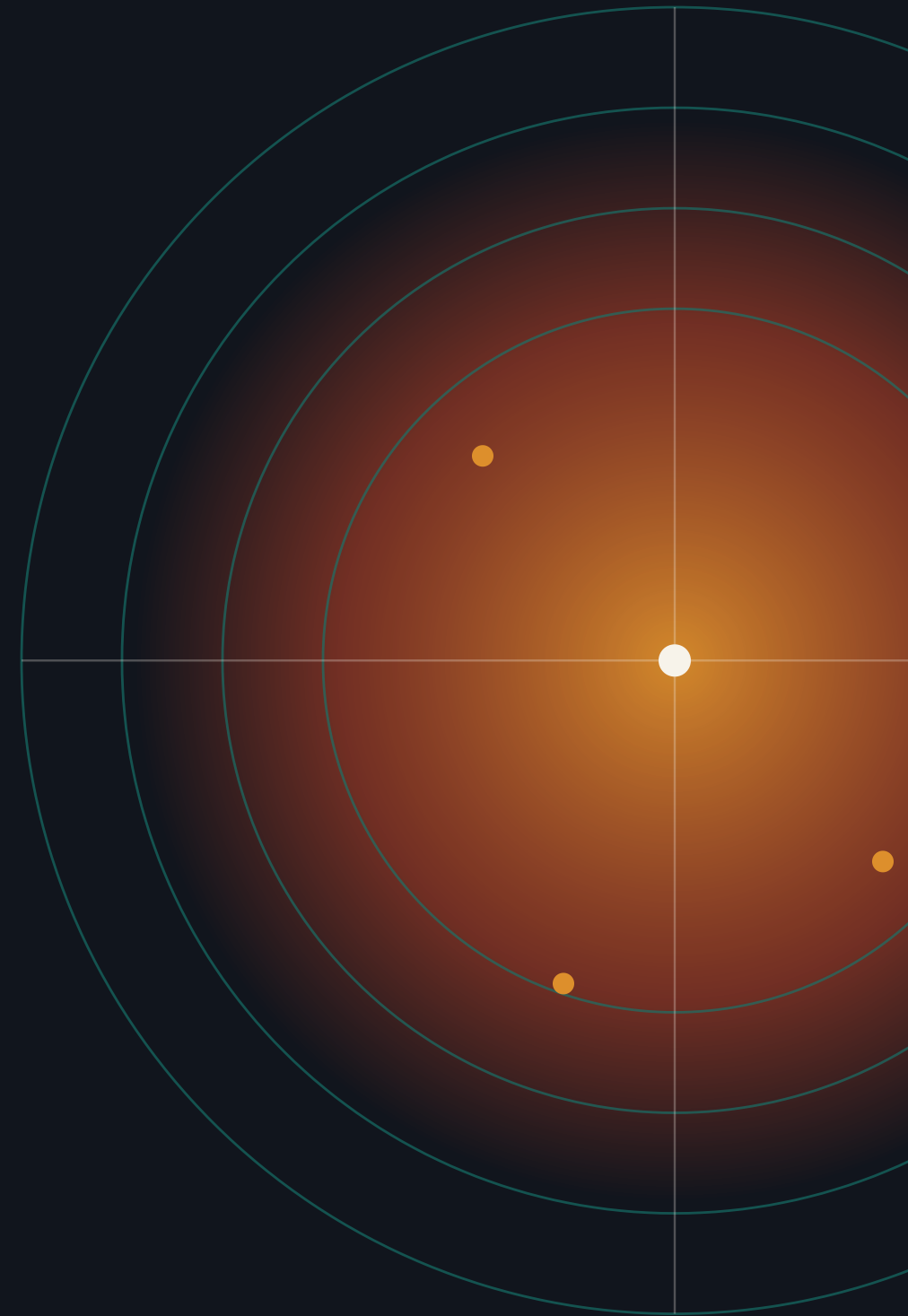
# How Much Local Data Is Enough?

An ontology-grounded neurosymbolic research agenda  
for transferability in forecasting

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# Brazil measures the sun in a *few places*.

- A handful of well-instrumented sites have years of clean records. **Vast regions have almost nothing.**
- The pragmatic hope is transfer learning to forecast **solar irradiation** [Pan & Yang 2010]: train where data is plentiful, leverage where it is not.

*But leveraging is not epistemically neutral.*

Leveraging presupposes the target is the *same kind of place* as the source.

TWO STATIONS MAY

share a latitude  
but differ in cloud regime

TWO STATIONS MAY

share a biome  
but differ in altitude

TWO STATIONS MAY

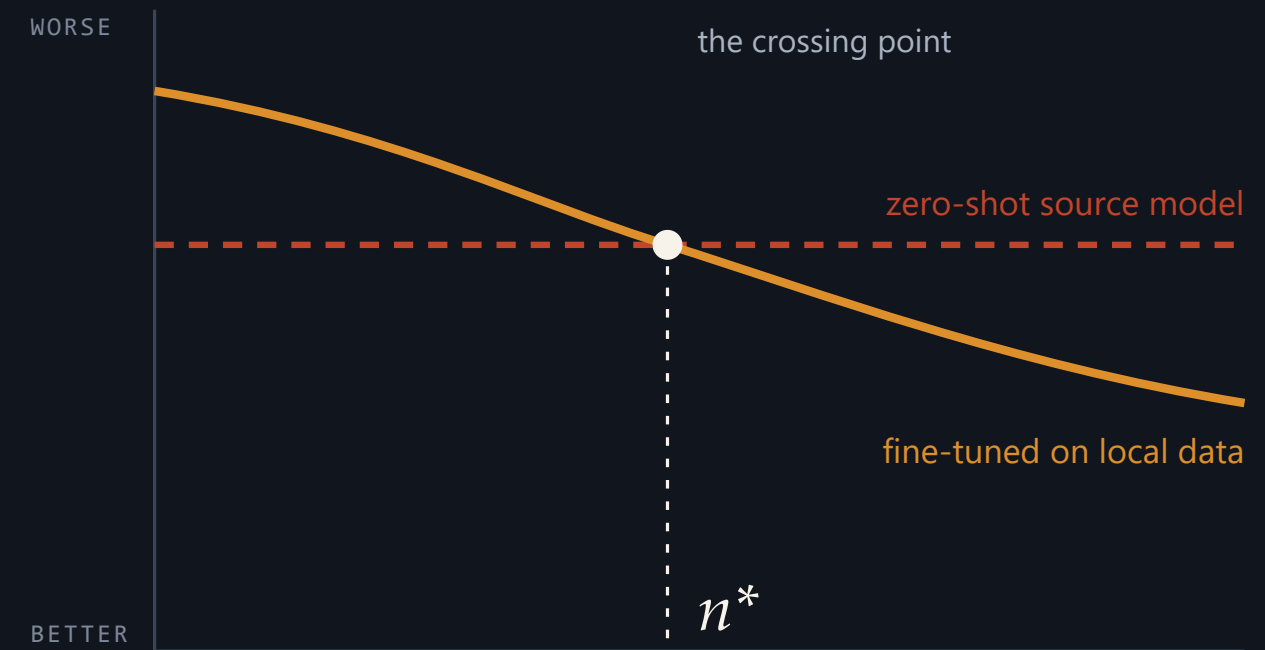
sit far apart  
yet behave identically

Diverge along the dimensions that matter and transfer **degrades performance rather than improving it** — negative transfer. Which similarities warrant transfer is a question **formal ontology** was built to clarify.

# How much local data is *enough*?

How much must be collected before **fine-tuning** beats  
deploying the source model as-is?

Call it the site's **data budget**,  $n^*$ .



# A network is mute about the scope of its own validity.

- Today the transfer decision is purely statistical. The network carries no account of **why** or **where** learned patterns should generalize. The neurosymbolic response is to couple the learner with an explicit symbolic layer that can be interrogated in human-legible terms [Garcez et al. 2023 · Kautz 2022].
- The promise is not only accuracy but **governance** — a model whose limits can be stated in plain terms, including where it should *not* be trusted.

*A symbolic gate is only as good as the **categories** it reasons with.*

Köppen climate types, biome schemes and bioclimatic distances **already are domain ontologies**: structured vocabularies for organizing the world. Grounding the gate in a **well-founded** solar-forecasting ontology [Guizzardi 2005] is what makes its commitments semantically precise, inspectable and **independently auditable** [Tiddi et al. 2022] — explicit knowledge **to improve transfer learning**, rather than an ad-hoc feature set.

# *A Neurosymbolic Semantic Gate* between the model and the place.

A symbolic decision layer, **still under development**, that would reason — in terms a domain scientist can read — about whether and how readily transfer is warranted.

IN

A trained source model  
and a candidate target site

THE GATE

A cascade of five  
symbolic criteria

OUT

A human-readable verdict  
and a predicted  $n^*$

APPROVED

WARNING

REJECTED

— a defeasible epistemic judgment, not a fact about the world.

Each criterion is a *distinct, testable hypothesis* about what licenses transfer.

| CRITERION               | NOTION OF “SAMENESS”                 | ONTOLOGICAL SOURCE           |
|-------------------------|--------------------------------------|------------------------------|
| Radiative compatibility | Comparable top-of-atmosphere forcing | Astronomy / site geometry    |
| Seasonal alignment      | Matching shape of the annual cycle   | Climatology                  |
| Climate identity        | Same Köppen class                    | Köppen-Geiger classification |
| Ecological proximity    | Near in biome / climate space        | Whittaker biome scheme       |
| Geographic nearness     | Small great-circle distance          | Spatial proximity            |

These are **distinct relations that often disagree**. Which notion of sameness, if any, actually carries predictive weight?

The gate will be measured against a *ground truth* that already exists.

20

BRAZILIAN SITES,  
BUDGET ALREADY  
ESTABLISHED

380

SOURCE-TARGET  
COMBINATIONS  
20 × 19, EVERY SITE A  
SOURCE

2,100

KM ACROSS BRAZIL

#### H1 — ORDERING

Sites judged **more similar** require **fewer local observations** to adapt — a measurably lower  $n^*$ .

#### H2 — DISCRIMINATION

The five criteria are **not interchangeable** — the empirical record will tell them apart.



THE 20 SITES ACROSS BRAZIL

Controlled for target-site difficulty, so similarity is not confused with a site simply being easier to forecast.

# Three overlapping work packages, *not* a sequence of phases.

## WP1

### Measure the budget

$n^*$  measured for all 20 sites under a fixed protocol — the ground-truth reference.

COMPLETE

## WP2

### Formalize & build the gate

Source and target as **relational roles**, the graded relations that mediate them, and the epistemic status of a verdict.

MOSTLY COMPLETE

## WP3

### Analyze against measured budgets

Test Ordering and Discrimination. The results **validate or falsify** the central claims.

TESTING PHASE

*Transferability is a structured, investigable phenomenon — and formal ontology has a concrete, testable role in explaining it.*

The same skeleton — a source model, a new context, a local adaptation budget, and competing similarity relations that may predict it — appears in hydrology, air quality, agriculture and almost every renewable source. The general contribution is a **way of asking**: state the ontological commitment, give it a role in a neurosymbolic system, and let the empirical outcome adjudicate.

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**PUSHBACK WANTED ON**

How I individuate and operationalize the similarity relations.

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**PUSHBACK WANTED ON**

The epistemic status I assign to a gate's verdict.

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**PUSHBACK WANTED ON**

How to frame the ontology–learning interface so it speaks to both communities.

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# Thank you.

Questions and feedback are welcome.



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